

Limitly - Software Requirements Specification (SRS)

1. Introduction

Limitly is a distributed rate limiting service designed to protect modern APIs from abuse. It provides configurable limits using API keys, endpoints, and optional user identifiers.

2. System Overview

The system acts as a middleware service that evaluates incoming API requests and determines whether they should be allowed or blocked based on defined rate limiting strategies.

3. Functional Requirements

3.1 Rate Limiting Engine: Implements Sliding Window Log algorithm using Redis.

3.2 API Endpoint: POST /check-limit to validate requests.

3.3 API Key Management: Each client gets a unique API key.

3.4 Rule Configuration: Limits per API key, endpoint, and user.

3.5 Analytics: Tracks total, allowed, and blocked requests.

3.6 Dashboard: Displays usage metrics, top users, and endpoint statistics.

4. Non-Functional Requirements

4.1 Performance: Response time should be under 100ms.

4.2 Scalability: Must support multiple clients and distributed architecture.

4.3 Reliability: Should handle failures gracefully.

4.4 Security: API keys must be securely managed.

5. System Design

Core Components:

- Rate Limiter Service (fast path)
- Analytics System (background processing)
- Redis for temporary and aggregated data storage
- Optional MongoDB for persistent storage

6. Data Design

Rate Limiter Key: rate_limit:{apiKey}:{endpoint}:{userId}

Analytics Key: analytics:top_users:{apiKey}:{date}

Uses TTL for automatic cleanup.

7. API Specification

POST /check-limit

Request: { apiKey, endpoint, userId(optional) }

Response: { allowed, remaining, retryAfter }

8. Constraints

System should avoid storing full logs to prevent memory issues.

Uses approximate analytics instead of exact logging.

9. Future Enhancements

Support for multiple strategies (Token Bucket, Fixed Window)

Payment integration (Pay-as-you-go model)

Multi-region deployment